Comparison of Depth Anything V2 Models and Speed-Quality Trade-off Analysis

1. Introduction

In this experiment, two different versions of the **Depth Anything V2** monocular depth estimation model were evaluated and compared:

- **Depth Anything V2 Small**
- **Depth Anything V2 Base**

The evaluation was performed using the **Depth8 dataset**, which contains only 8 RGB-D image pairs. Therefore, all available images in the dataset were used for inference and comparison.

The objective of this experiment was to analyze the trade-off between **depth estimation quality, computational cost, memory consumption, and processing speed**.

2. Evaluation Setup

For each model, inference was performed on all 8 RGB images. The following factors were measured:

- Depth estimation quality
- GPU memory consumption
- Average inference time
- Preprocessing time
- Postprocessing time
- Total processing time
- FPS (Frames Per Second)
- Input image size

Since Depth Anything V2 predicts **relative depth**, while the Depth8 ground-truth maps contain **metric depth values**, the predicted depth maps were first aligned with the ground truth using a scale-and-shift transformation:

$D_{aligned} = aD_{pred}+b$

where a is a scale factor and b is a shift value estimated using least squares optimization.

After alignment, the following metrics were calculated:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Error)
- AbsRel (Absolute Relative Error)

3. Results

The average results over the 8 images are shown below:

Metric	Depth Anything V2 Small	Depth Anything V2 Base
Number of images	8	8
MAE (m)	0.1042	0.0985
RMSE (m)	0.1311	0.1247
AbsRel	0.0672	0.0634
Preprocessing time (ms)	21.71	19.32
Inference time (ms)	78.58	125.80
Postprocessing time (ms)	40.79	40.58
Total processing time (ms)	141.08	185.70
FPS	7.51	5.91
Peak GPU Memory (MB)	257.40	628.69
Input Size	392 × 518	392 × 518

4. Quality Comparison

The Depth Anything V2 Base model achieved slightly better depth estimation accuracy compared with the Small model.

The error values were:

MAE

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|D_{aligned}^{(i)} - D_{gt}^{(i)}|$$

- Small: **0.1042 m -** This means that, on average, the predicted depth differs from the ground truth by approximately 10.4 cm.
- Base: **0.0985 m**

RMSE

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(D_{aligned}^{(i)} - D_{gt}^{(i)})^2}$$

- Small: **0.1311 m**
- Base: **0.1247 m**

AbsRel

$$AbsRel = \frac{1}{N} \sum_{i=1}^{N} \frac{|D_{aligned}^{(i)} - D_{gt}^{(i)}|}{D_{gt}^{(i)}}$$

- Small: **0.0672**
- Base: **0.0634**

The Base model reduced the AbsRel error from 0.0672 to 0.0634, showing a small improvement in depth prediction accuracy.

However, the improvement is relatively limited compared with the additional computational cost required by the Base model.

5. Speed and Computational Cost Analysis

Preprocessing Time

Preprocessing includes:

- image conversion,
- resizing,
- normalization,
- tensor generation,
- transferring data to the GPU.

The preprocessing times were similar for both models:

- Small: **21.71 ms**
- Base: **19.32 ms**

This similarity is expected because both models use the same preprocessing pipeline.

Inference Time

Inference time represents the time required for the neural network forward pass.

The results were:

- Small: **78.58 ms**
- Base: **125.80 ms**

The Base model required significantly more computation because it has a larger architecture with more parameters.

Although the Base model provides slightly better depth accuracy, this improvement comes at the cost of approximately 60% longer inference time.

Postprocessing Time

Postprocessing includes:

- resizing the predicted depth map,
- converting the output tensor to NumPy format,
- aligning depth values with ground truth,
- calculating evaluation metrics.

The postprocessing times were almost identical:

- Small: **40.79 ms**
- Base: **40.58 ms**

This indicates that postprocessing is mostly independent of the model architecture.

6. FPS Analysis

FPS (Frames Per Second) indicates how many images can be processed in one second.

It is calculated as:

$$FPS = \frac{1}{T_{total}}$$

$$T_{total} = T_{preprocessing} + T_{inference} + T_{postprocessing}$$

where T_{total} is the total processing time in seconds.

For the Small model: FPS = 1 / 0.14108 = 7.51

For the Base model: FPS = 1 / 0.18570 = 5.91

The results show:

Model	FPS
Depth Anything V2 Small	7.51
Depth Anything V2 Base	5.91

The Small model processes more frames per second because it requires less computation.

7. 20 FPS Requirement Analysis

The system requirement is: FPS ≥ 20

A 20 FPS system must process each frame within:

$T_{max} = 1 / 20 = 0.05s = 50ms$

The measured results are:

Model	Total Time	FPS	Meets 20 FPS?
Depth Anything V2 Small	141.08 ms	7.51	No
Depth Anything V2 Base	185.70 ms	5.91	No

Therefore, neither model satisfies the 20 FPS real-time requirement on the tested hardware.

8. Model Selection and Trade-off Discussion

The Base model provides slightly better depth estimation accuracy, with lower MAE, RMSE, and AbsRel values. However, this improvement requires:

- higher inference time,
- lower FPS,
- significantly higher GPU memory consumption.

The Small model provides a better balance between accuracy and computational efficiency:

- Faster processing (7.51 FPS vs 5.91 FPS)
- Lower GPU memory usage (257 MB vs 629 MB)
- Similar depth estimation quality

Therefore, if a choice must be made between these two models, **Depth Anything V2 Small is the more practical option for applications where speed and memory are important**.

However, for a system that strictly requires **at least 20 FPS**, neither model is suitable without further optimization. Possible solutions include:

- using a smaller model variant,
- reducing input resolution,
- using quantization methods,
- using more powerful hardware.

9. Conclusion

The experiment demonstrated the trade-off between model accuracy and computational efficiency in monocular depth estimation.

Depth Anything V2 Base achieved slightly better depth accuracy but required substantially more computation and memory. Depth Anything V2 Small achieved lower latency and better efficiency with only a small decrease in accuracy.

For real-time applications, model selection should consider not only accuracy but also speed, memory limitations, and hardware constraints. In this experiment, the Small model provides a better efficiency-quality balance, although additional optimization would be required to achieve a 20 FPS real-time system.